Reward function

$$r_t = -\left(w_1 x_{1,t} + w_2 x_{2,t} + w_3 x_{3,t} + w_4 \left(1 - \cos \psi_t\right) + w_5 x_{5,t}\right) \tag{1}$$

where, for decision step t (5 s):

- $x_{1,t} \in \{0, 1\}$ is 1 if the aircraft has a loss of separation (< 5 NM), with $w_1 = 100$;
- $x_{2,t} \in \{0, 1\}$ is 1 if the aircraft is inside a restricted area, with $w_2 = 10$;
- $x_{3,t} \in \{0, 1\}$ is 1 if the aircraft is outside the sector, with $w_3 = 10$;
- ψ_t is the angle between the aircraft's heading and the bearing to its waypoint, with $w_4 = 1$;
- $x_{5,t} = 1$ is a time penalty for every decision step, with $w_5 = 0.1$.

The term $1 - \cos \psi_t$ is 0 when flying straight at the waypoint and 2 when flying away from it. The time penalty makes a faster flight to the waypoint better. The reward is never positive.